PG No. 17　$C_{3\mathrm{i}}$　$\bar{3}$　[trigonal]

* character table

$C_{3\mathrm{i}}$	$1(1)$	$3^+_{001}(2)$	$-1(1)$	$-3^+_{001}(2)$
A_g	1	1	1	1
E_g	2	-1	2	-1
A_u	1	1	-1	-1
E_u	2	-1	-2	1

* polar $\leftrightarrow$ axial conversion

A_g (A_u)　E_g (E_u)　A_u (A_g)　E_u (E_g)

* symmetric product

	A_g	E_g	A_u	E_u
A_g	A_g	E_g	A_u	E_u
E_g		$A_g + E_g$	E_u	$2A_u + E_u$
A_u			A_g	E_g
E_u				$A_g + E_g$

* anti-symmetric product

A_g	E_g	A_u	E_u
$-$	A_g	$-$	A_g